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Surrogate model of DEM simulation for binary-sized particle mixing and segregation

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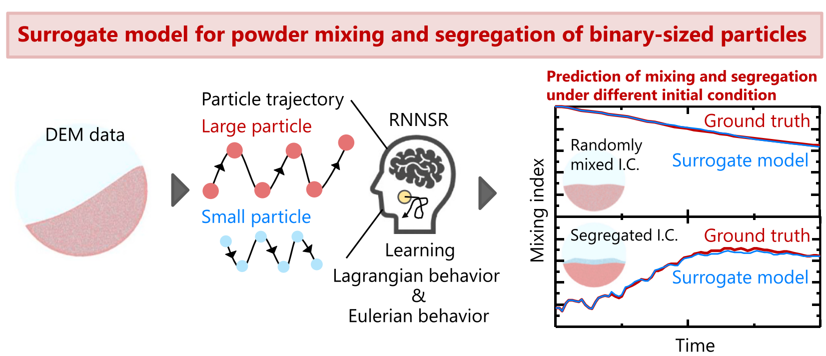
HIGHLIGHTS

- Surrogate model of DEM simulation for powder mixing, namely RNNSR, was developed.
- RNNSR was extended by adding particle size data as a feature into the training data.
- Segregation phenomenon was accurately predicted by extended-RNNSR.
- Extended-RNNSR was applicable to predicting segregation under different conditions.
- Extended-RNNSR enabled ultra-fast computing even in binary-sized particle systems.

GRAPHICAL ABSTRACT



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ABSTRACT

Segregation is a well-known phenomenon that occurs during powder mixing, wherein particles with similar properties are collected from specific regions of a bulk powder. Numerical simulation using the discrete element method (DEM) is recognized as a potent tool for investigating and predicting segregation. However, DEM simulations are computationally expensive. To address the high computational overhead, in our prior work, we proposed a machine-learning-based surrogate model for DEM simulation, namely, recurrent neural network with stochastically calculated random motion (RNNSR). The model was designed to predict the local mean component of the particle behavior using the Lagrangian approach and the local variability component of the particle behavior using the Eulerian approach. However, the RNNSR was demonstrated exclusively for monodisperse particles with homogeneous properties. Hence, in the current study, we extended the RNNSR to simulate the mixing and segregation of powders with inhomogeneous properties. The dependence of the particle size on the Lagrangian and Eulerian behaviors of the particles was investigated. Based on this analysis, an extended-RNNSR was developed for binary-sized particle system by adding the particle size data for the training data. The prediction accuracy of the extended-RNNSR was evaluated in terms of the mixing degree, particle velocity, granular temperature, and computing speed. It was demonstrated that the extended-RNNSR constructed on learning data initiated from the randomly mixed initial condition could predict mixing and segregation initiated from both

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segregated as well as the randomly mixed initial conditions. The extended-RNNSR also demonstrated a much faster computing speed than the DEM.

1. Introduction

Powder mixing is a fundamental operation in various industrial sectors, including chemicals, pharmaceuticals, cosmetics, batteries, agriculture, and food. Segregation is a well-known phenomenon that can occur during powder mixing, wherein particles with similar properties (e.g., size, density, and shape) are collected in specific regions within a bulk powder, resulting in the unintended separation of particles [1]. Segregation is generally regarded as an undesirable phenomenon in the industrial sector because it adversely affects the homogeneity of the target powder and final products. Therefore, understanding and predicting segregation are critical. Numerical simulation using the discrete element method (DEM) is considered an effective approach for investigating and predicting segregation [2]. DEM allows for a comprehensive,

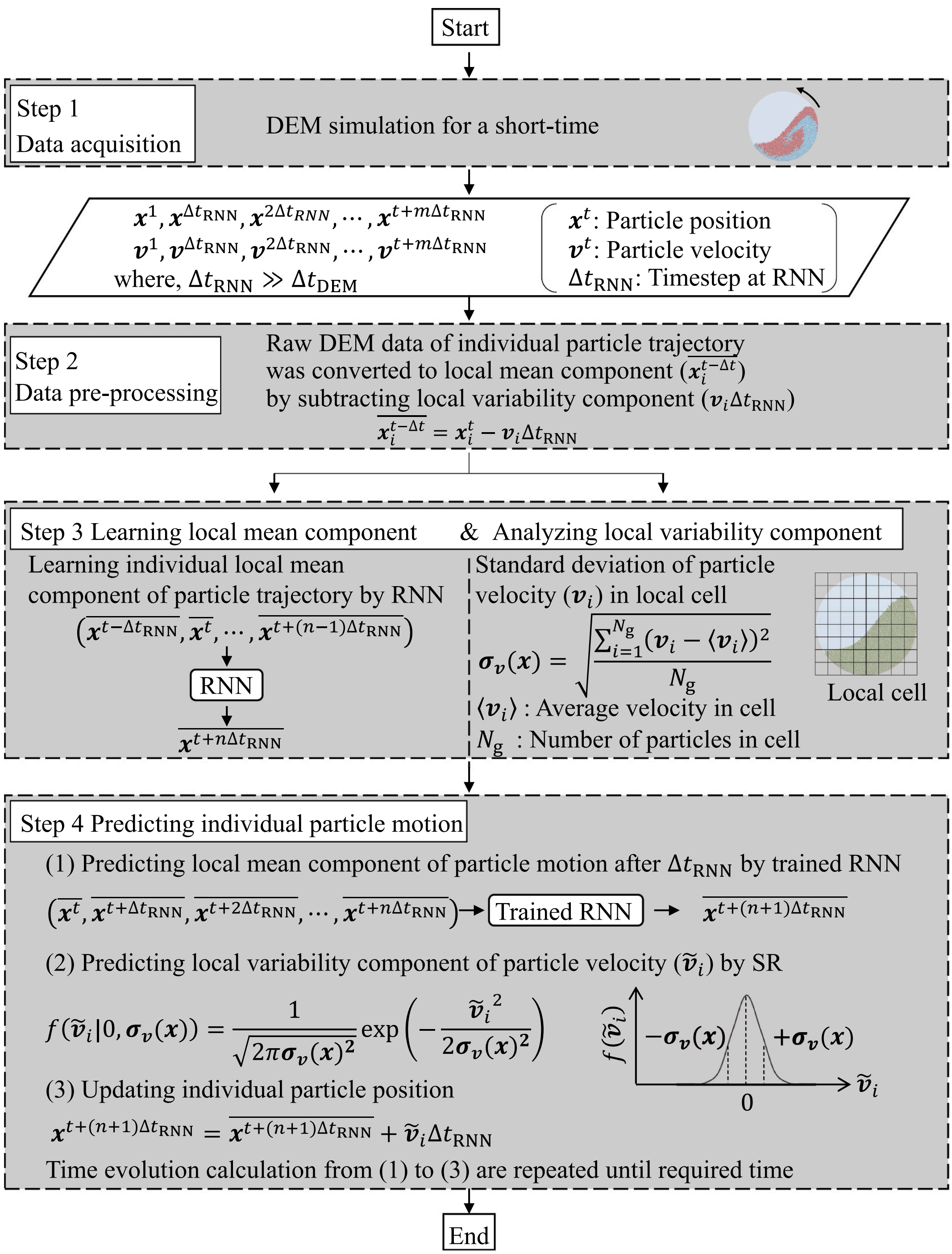


Fig. 1. Calculation procedure of the recurrent neural network with stochastically calculated random motion (RNNSR) for monodispersed particles with homogeneous particle size.

particle-level investigation of the segregation phenomenon through the modeling of any mixture of particles with inhomogeneous sizes, densities, and shapes. However, DEM simulations are computationally expensive, which limits their application to systems with smaller spatiotemporal scales.

In a DEM simulation, the particle behavior is determined by iteratively solving the equations of motion for each particle. This computation involves two iterative processes based on the number of particles and time evolution [3]. Thus, DEM simulations with a large number of particles and long-term evolution require extremely long computing times. For example, simulating a large quantity of powder, over a billion particles, and powder behavior over several minutes, which are typical spatiotemporal scales encountered in actual powder-mixing processes, is a significant challenge. Although advancements in hardware have extended the number of particles and time scale that can be handled in DEM simulations, these capabilities are still limited to tens of millions of particles and several seconds [4,5]. This substantial gap between the DEM and actual processes emphasizes the need for developing methods to enable DEM simulations to handle long-term powder mixing and segregation involving a large number of particles.

Coarse-grained (CG) methods have been investigated to address the high computational overhead associated with a large number of particles. In a CG method, artificially scaled-up particles are used to reduce the total number of particles, thereby enabling larger-scale simulations. Suitable scaling laws are required to accurately represent the original particle behavior using scaled-up particles. Various scaling laws have been proposed for multiphase flows [6–10], cohesive powders [11–15], and nonspherical particles [14]. Scaling laws for bi- and polydisperse particles [14,16–18] have also been proposed, and CG-DEM simulations of powder segregation have been reported [18]. For powder mixing processes, we developed a CG method for contact-dominant granular shear flow and demonstrated its effectiveness in a rotating drum mixer and vertical high-shear mixer [19–21]. The limitations associated with the number of particles can be overcome by using the CG method.

On the other hand, alternative approaches have been developed to address the high computational cost associated with long-term simulations. DEM extrapolation methods using bijective functions have been proposed to predict the particle behavior and residence time in continuous processes [22,23]. Additionally, a reduced-order model based on proper orthogonal decomposition (POD) demonstrated the low-cost prediction of multiphase flow [24]. Recently, machine learning (ML)-based surrogate models have been developed for high-speed computing. These models can be categorized into two types based on the predictor variables. The first type could predict bulk properties such as the mass discharge rate [25–28], degree of mixing [29,30], and dynamic angle of response [31] from the microscopic particle properties and operating parameters using an ML model without running full DEM simulations. These ML models were pretrained on input data datasets and the corresponding bulk powder properties obtained from DEM simulations. However, these methods cannot predict the behavior of individual particles. In contrast, the second type of ML-models directly predicted the individual particle behavior, similar to the DEM simulation results. For this type, various ML models such as graph neural networks [32,33] and convolutional neural networks [34,35] were employed. Recently, we proposed an ML-based surrogate model, namely, recurrent neural network with stochastically calculated random motion (RNNSR) [36,37]. The RNNSR was designed to learn particle dynamics from short-term DEM simulation results and predict convective and diffusive mixing by combining an RNN with a stochastic random model. The RNNSR learns and predicts individual particle trajectories with a significantly larger time step than that used in the DEM, enabling ultrafast computation of the time evolution of particle behavior. The accuracy of the RNNSR was demonstrated for powder mixing in terms of the degree of mixing, particle velocity, granular temperature, and computing speed [36,37]. However, a common limitation of existing ML-based surrogate models, including RNNSR, is that they have been demonstrated exclusively for monodisperse particles with homogeneous properties. However, an ML-based surrogate model capable of simulating the dynamics of particles with inhomogeneous properties and predicting mixing and segregation has not been reported heretofore. Addressing this issue is crucial to extend the capabilities of ML-based surrogate models for powder simulations.

The objective of this study was to develop an ML-based surrogate model to simulate the mixing and segregation of powders with inhomogeneous properties. This study focused on a binary-sized particle system and developed an extended-RNNSR, which is capable of predicting segregation. The RNNSR was originally designed to predict the local mean component of particle behavior using the Lagrangian approach and the local variability component of particle behavior using the Eulerian approach [36,37]. Thus, to gain insight into extending the RNNSR to binary-sized particle systems, we first conducted a DEM analysis of the Lagrangian and Eulerian behaviors of binary-sized particles during segregation in a rotating drum mixer. The dependence of the particle size on the Lagrangian and Eulerian particle behaviors was investigated. Based on this analysis, an extended-RNNSR for binary-sized particle system was developed. Finally, using the extended-RNNSR, the mixing and segregation of binary-sized particles in a rotating drum were predicted. The prediction accuracy of the extended-RNNSR was evaluated in terms of the mixing degree, particle velocity, granular temperature, and computing speed. Furthermore, we investigated the effect of the initial condition of the binary-sized particles in the training data of the RNNSR on its prediction accuracy.

2. Method

2.1. RNNSR [36,37]

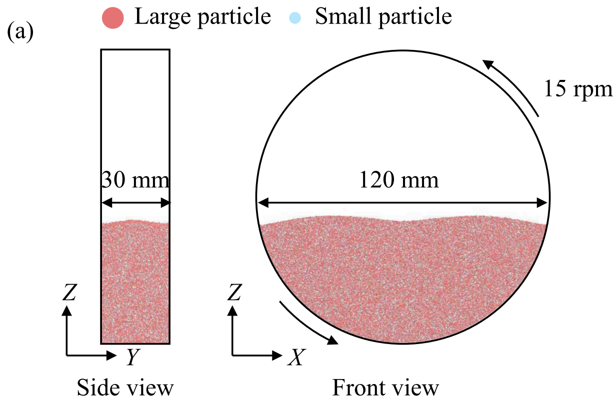
Calculation procedure of the RNNSR for monodisperse particles with homogeneous particle size [36,37] is outlined in Fig. 1. In our prior study, we determined individual particle behavior using RNNSR in four steps. First (step 1), a short-term DEM simulation was performed to acquire training data for constructing the RNNSR. The governing equations of the DEM are presented in the Appendix. In this step, the DEM simulation was performed for a significantly shorter duration than the target prediction time. Thereafter, time series of the individual particle trajectory and particle velocity every Δt_{RNN} intervals were extracted from the DEM simulation results. The Δt_{RNN} was significantly larger than the time step used in the DEM.

In the next step (step 2), the data were preprocessed to prepare training data for the RNN [36,37]. The RNNSR employs an RNN to learn and predict the local mean components of a particle trajectory. Thus, the raw DEM data were needed to be converted into local mean components [36,37]. We used Eq. (1) to convert the raw DEM data into RNN training data.

$$\overline{\boldsymbol{x}_i^{t-\Delta t}} = \boldsymbol{x}_i^t - \boldsymbol{v}_i \Delta t_{\mathrm{RNN}} \tag{1}$$

where the original particle position ($\boldsymbol{x}_i^t$) was converted into the local mean component of particle position ($\overline{\boldsymbol{x}_i^{t-\Delta t}}$) by subtracting displacement derived from the original particle velocity ($\boldsymbol{v}_i \Delta t_{\mathrm{RNN}}$) [36,37]. $\boldsymbol{v}_i \Delta t$ means the local variability component of particle position. $\boldsymbol{v}_i$ denotes the particle velocity at the current step.

In the third step (step 3), the local mean component was learned using an RNN, and the local variability component was analyzed. In this step, the RNN learned the local mean components of the individual particle trajectories. Simultaneously, the calculation of standard deviation of particle velocity in a local space ($\boldsymbol{\sigma_v}(\boldsymbol{x})$) was performed. The simulation system was divided into fixed cubic cells, and $\boldsymbol{\sigma_v}(\boldsymbol{x})$ was calculated for each cell. $\boldsymbol{\sigma_v}(\boldsymbol{x})$ was defined by the following equation [36,37]:



(b)

Particle diameter	[mm]	0.5, 0.25
Time step	[μs]	2.0
Particle density	[kg/m^3]	2500
Young modulus	[Pa]	1.0×10^7
Poisson ratio	[-]	0.3
Restitution coefficient	[-]	0.75
Friction coefficient	[-]	0.575
Rolling friction coefficient	[-]	0.01
Number of particles	[-]	1,000,000 & 2,000,000

Fig. 2. (a) Configuration of the rotating drum mixer simulated in this study and initial condition of the particles. Large particles are displayed in red, and small particles are displayed in cyan. The particle sizes were set to induce segregation. (b) Calculation conditions for the DEM simulation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x}) = \sqrt{\frac{\sum_{i=1}^{N_g}(\boldsymbol{v}_i - \langle \boldsymbol{v}_i \rangle)^2}{N_g}} \tag{2}$$

where $\boldsymbol{v}_i$, $\langle \boldsymbol{v}_i \rangle$, and N_g denote the particle velocity of i-th particle in the local cell, mean particle velocity in the local cell, and number of particles in the local cell, respectively. $\boldsymbol{x}$ represents the coordinates of the local cell. $\langle \boldsymbol{v}_i \rangle$ was calculated using spatiotemporal averaging. $\boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x})$ denotes a vector quantity, with each component corresponding to the variance of the respective velocity component. $\boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x})$ was determined only for the cells with sufficient number of particles, while zero was set for the other cells with much less particles.

In the final step (step 4), the individual particle behavior was predicted using a trained RNN and a stochastic random (SR) model. First, the local mean component of individual particle position after Δt_{RNN} ($\overline{\boldsymbol{x}^{t+(n+1)\Delta t_{\mathrm{RNN}}}}$) was predicted by the trained RNN. Next, the local variability component of particle velocity ($\widetilde{\boldsymbol{v}}_i$) was estimated using the SR model based on the following Gaussian distribution:

$$f\left(\widetilde{\boldsymbol{v}}_i|0, \boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x})\right) = \frac{1}{\sqrt{2\pi\boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x})^2}} exp\left(-\frac{\widetilde{\boldsymbol{v}}_i^{\,2}}{2\boldsymbol{\sigma}_{\boldsymbol{v}}(\boldsymbol{x})^2}\right) \tag{3}$$

Finally, the time evolution of each particle trajectory was calculated using the local mean component ($\overline{\boldsymbol{x}^{t+(n+1)\Delta t_{\mathrm{RNN}}}}$) predicted by the trained RNN and the local variability component ($\widetilde{\boldsymbol{v}}_i$) estimated by the SR model.

$$\boldsymbol{x}^{t+(n+1)\Delta t_{\mathrm{RNN}}} = \overline{\boldsymbol{x}^{t+(n+1)\Delta t_{\mathrm{RNN}}}} + \widetilde{\boldsymbol{v}}_i \Delta t_{\mathrm{RNN}} \tag{4}$$

The predicted $\boldsymbol{x}^{t+(n+1)\Delta t_{\mathrm{RNN}}}$ was repeatedly input into the RNN and SR models until the target prediction time.

In the RNNSR, Lagrangian and Eulerian behaviors were combined. The local mean component of the individual particle trajectories was predicted in a Lagrangian manner, whereas the local variability

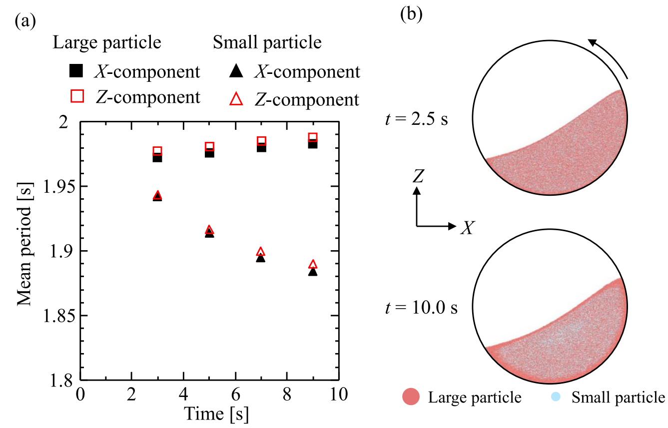


Fig. 3. (a) Mean period of individual particle trajectories obtained from FFT analysis for large and small particles as a function of time during segregation. (b) Snapshots of the particles at 2.5 and 10.0 s. Large particles are displayed in red, and small particles are displayed in cyan. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

component in a fixed local space was predicted in an Eulerian manner. To apply the RNNSR to a binary-sized particle system, it was deemed necessary to analyze how the particle size affects the Lagrangian and Eulerian behavior. Therefore, we performed a DEM analysis of the Lagrangian and Eulerian behaviors of the binary-sized particles.

2.2. DEM analysis of binary-sized particle behavior

2.2.1. Calculation conditions

DEM analysis of the Lagrangian and Eulerian behaviors of binary-sized particles was performed to investigate the particle size dependency on the Lagrangian–Eulerian behavior. Fig. 2 shows the calculation conditions for the DEM analysis. A three-dimensional horizontal rotating drum mixer was used. The rotating drum consisted of a horizontal cylindrical vessel with a diameter of 120 mm and depth of 30 mm. In the study, binary-sized spherical particles were used. Here, large particles (diameter 0.5 mm) are displayed in red, and small particles (0.25 mm in diameter) are displayed in cyan. The particle sizes were set to induce segregation. The initial condition of the DEM simulation was set to a randomly mixed condition of large and small particles. The input parameters for the DEM simulation were determined based on previous DEM studies on binary-sized particle behavior in rotating drum mixers [38–40]. The number of large and small particles was 1 million and 2 million, respectively. A powder mixing simulation was conducted for 10 s, and the particle behavior was analyzed in terms of Lagrangian and Eulerian behaviors. All the DEM simulations were performed using EDEM 2023 (Altair Engineering Inc.).

2.2.2. Effects of particle size on Lagrangian behavior

To examine the dependence of particle size on Lagrangian behavior, individual particle trajectories were analyzed. The RNNSR learns and predicts the particle trajectory, which is the time-series data of the particle positions. In batch-type mixing processes, including rotating drum mixers, particle trajectories exhibit periodic changes over time; thus, the period is an important characteristic. Accordingly, we examined the Lagrangian behavior by analyzing the period of the particle trajectories for each particle size. The RNNSR was modeled to learn and predict particle trajectories under a steady state; therefore, we preliminary identify the steady state. The temporal change in the particle kinetic energy over time computed by the DEM was shown in Supplementary material (Fig. S1). The kinetic energy reached a steady state after 2 s. Thus, the particle trajectories were extracted for 2 s after the particle kinetic energy reached a steady state, and their periods were

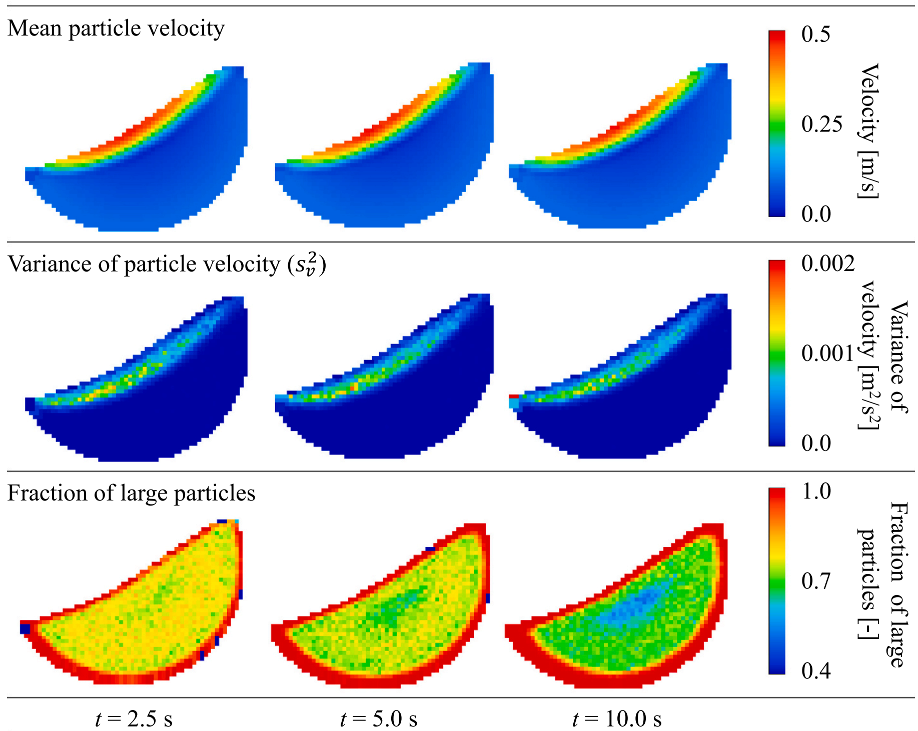


Fig. 4. Contour maps of instantaneous mean particle velocity, variance of particle velocity, and fraction of large particles within fixed cells at different times during segregation.

obtained using fast Fourier transformation (FFT) analysis. The periods of the particle trajectory were obtained for the X- and Z-components perpendicular to the rotation axis. The periods for all the particles were calculated and averaged separately for large and small particles. Fig. 3a shows the mean period of individual particle trajectories for large and small particles. The results showed that the mean period increased for large particles and decreased for small particles over time. The increase/decrease in the mean period for large/small particles can be explained by the segregation phenomenon simulated in this study. Fig. 3b shows photographs of the particles at 2.5 and 10.0 s. From 2.5 to 10.0 s, large particles tended to move toward the periphery region of the powder bed, whereas small particles migrated toward the central core of powder bed. It has been reported that this segregation phenomenon is due to particle size segregation caused by percolation [38]. The change in the particle spatial distribution due to the segregation lead to longer/shorter traveling distance per circulation for large/small particles, resulting in an increase/decrease in the period for large/small particles shown in Fig. 3a. These results indicate that the individual particle trajectories (Lagrangian behavior) were significantly affected by particle size.

2.2.3. *Effects of particle size on Eulerian particle behavior*

To investigate the particle size dependency on the Eulerian behavior, the mixer was divided into cubic cells. The cell size was set to four times the diameter of the large particles. The mean velocity, deviation of the particle velocity, and fraction of large particles within each cell were calculated. The variance in the particle velocity (s_v^2) was calculated using the following equation:

$$s_v^2 = \frac{\sum_{i=1}^{N_g} (|\boldsymbol{v}_i| - \langle|\boldsymbol{v}_i|\rangle)^2}{N_g} \tag{5}$$

where $|\boldsymbol{v}_i|$, $\langle|\boldsymbol{v}_i|\rangle$, and N_g represent the absolute particle velocity of i-th particle in the local cell, the mean absolute particle velocity in the local cell, and the number of particles in the local cell, respectively. Fig. 4 shows the instantaneous results at 2.5, 5.0, and 10.0 s. Owing to segregation, the fraction of large particles within each cell changed significantly over time. In contrast, the local mean velocity and variance of particle velocity did not change. This result indicated that despite the ongoing segregation, the locally averaged particle behavior remained unchanged. Consequently, we found that Eulerian behavior, which corresponds to the averaged behavior of particle assemblies within a fixed local space, was insensitive to particle size, even when segregation was in progress.

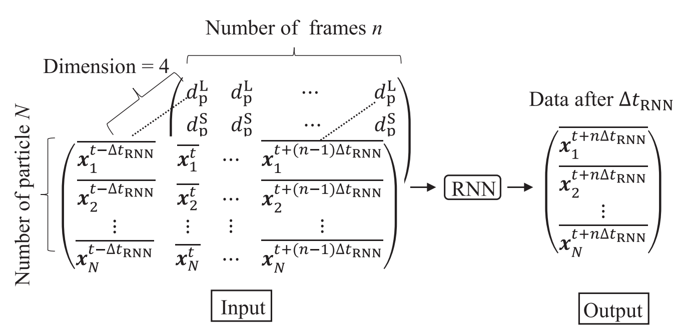


Fig. 5. Input and output data at the RNN-learning step in the extended-RNNSR. The particle size data was added to input data for the RNN. The data input to the RNN was a matrix consisting of the local mean components of particle trajectory and particle size over the number of particles N over an arbitrary number of frames n. d_p^L and d_p^S is large and small particle size-data, respectively. The output data was the local mean component of particle positions after Δt_{RNN}.

2.3. *Extended-RNNSR for binary-sized particles*

DEM analyses of the Lagrangian and Eulerian behaviors revealed that the Lagrangian behavior was dependent on the particle size, whereas the Eulerian behavior was independent of the particle size. These findings suggested that an RNN should be designed to predict the local mean components of individual particles in a binary-sized system by accounting for the particle size. However, the SR model was not

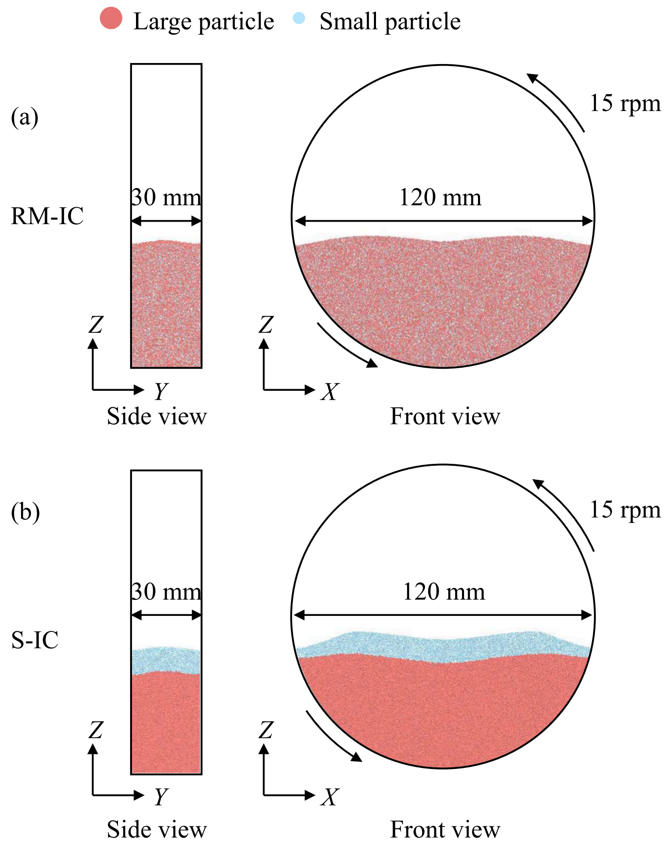


Fig. 6. Configuration of the rotating drum mixer simulated in this study and initial particle position. (a) Randomly mixed initial condition (RM-IC) and (b) segregated initial condition. (s = IC). Large and small particles are displayed in red and cyan, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Structure of the RNN model used in this study.

Layer type	Activation function	Number of parameters
GRU	–	1560
Dense	Hyperbolic tangent function	315
Dense	Linear	48
Number of frames of input data, *n*		15
Optimization method		Adam (learning rate = 0.01)
Initial value		He
Loss function		Mean squared error (MSE)
Epoch		20
Batch size		500

needed to be designed to predict the local variability component in a binary-sized system by considering the particle size. Based on these insights, an extended-RNNSR was developed for binary-sized particles. Specifically, the surrogate model for the local mean and local variability components was designed as follows:

1) To learn and predict the local mean components of individual particle trajectories (i.e., Lagrangian behavior), particle size data were added to the input data for the RNN, as shown in Fig. 5.
2) For the Eulerian analysis and the prediction of local variability components in a local cell, we did not differentiate between large and small particles. Specifically, in Step 3 in Fig. 1, the standard deviation of particle velocity in each local cell ($\boldsymbol{\sigma_v}(\boldsymbol{x})$) was calculated regardless of the particle size.

Table 2
Initial particle positions used in each step.

	Surrogate modeling	Prediction
Case 1	Randomly mixed	Randomly mixed
Case 2	Randomly mixed	Segregated
Case 3	Segregated	Randomly mixed
Case 4	Segregated	Segregated

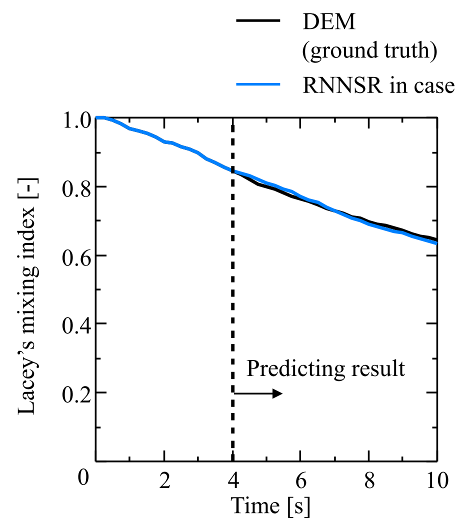


Fig. 7. Temporal changes in the degree of the particle mixing calculated by the DEM and RNNSR in the case 1. The data initiated from the RM-IC were employed in both the surrogate modeling and prediction step in the RNNSR.

3. Calculation conditions

To verify the extended-RNNSR, we simulated the mixing and segregation of binary-sized particles in a rotating drum. Fig. 6**a** and **b** show the initial conditions. All particles were stationary under the initial conditions. Two types of initial particle positions of binary-sized particles, which were (a) randomly mixed initial condition (RM-IC) and (b) segregated initial condition (S-IC) along the *Z*-direction, were employed to investigate the effect of initial condition on prediction accuracy. The DEM simulations of powder mixing and segregation were performed for 10 s starting from (a) the RM-IC and for 20 s starting from (b) the S-IC. The DEM simulation results were compared with the prediction results obtained using the extended-RNNSR. The calculation conditions for the DEM simulation were the same as those described in Section 2.2 (Fig. 2b).

The extended-RNNSR was implemented in Python using TensorFlow 2.13. Table 1 lists the structures of the RNN models used in the study. Typical settings were employed for the RNN training. In this study, time step used in RNN (Δt_{RNN}) was set to 0.05 s. Δt_{RNN} was optimized according to a criterion reported in a previous study [37]. The raw DEM simulation data were extracted every 0.05 s between 3.0 and 4.0 s (20 frames), in which the quasi-steady state of particle kinetic energy was ensured. Subsequently, the time-series data for 15 frames were input into the RNN. The results of RNN training are shown in the supplementary material (Fig. S2). The loss values in the training and test sets were sufficiently low, confirming that the RNN successfully learned the local mean components of the particle trajectory. A length of local cell size used to define the standard deviation of the particle velocity $\boldsymbol{\sigma_v}(\boldsymbol{x})$ was fixed at four times the diameter of the large particle. The cell size was optimized in a previous study [36].

To investigate the effect of the initial condition on the prediction accuracy of the extended-RNNSR, two types of initial particle positions were used in each step of surrogate modeling and prediction. Table 2

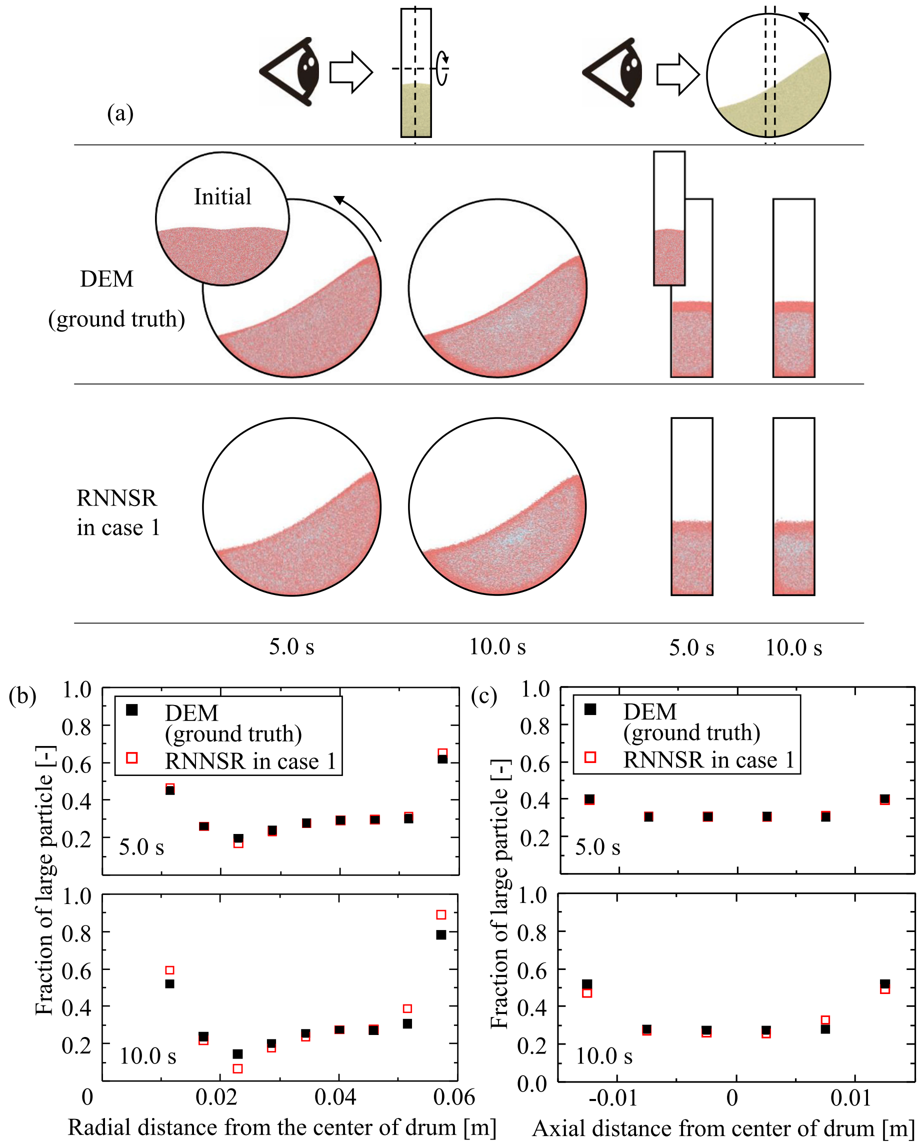


Fig. 8. (a) Powder mixing behavior on the cross-sections at the center of the drum from front and side views. (b) Fraction of large particles as a function of radial distance from the center of drum. (c) Fraction of large particles as a function of axial distance from the center of drum. The data initiated from the RM-IC were employed in both the surrogate modeling and prediction step in the RNNSR.

lists the initial particle positions used in each step. The initial particle position in the prediction corresponds to the target initial condition in the prediction.

4. Results and discussion

4.1. Performance of the extended-RNNSR at randomly mixed initial condition

The data initiated from the RM-IC were employed in both the surrogate modeling and prediction steps (case 1 in Table 2). Fig. 7 shows the Lacey's mixing index predicted by the RNNSR. The results from 4 to 10 s were predicted using the RNNSR. The Lacey's mixing index was used to measure the degree of powder mixing. Details of Lacey's mixing index can be found in previous studies [41]. Briefly, the drum mixer was divided into cubic cells, and large particles were used as tracers. In this study, the length of the cubic cell was set to 0.04*D*, where *D* denotes the drum diameter. We preliminarily performed a sensitive analysis of cell lengths (as shown in Supplemental material, Fig. S3) and confirmed that the change in Lacey's mixing index was almost negligible at cell lengths smaller than 0.04*D*. The result showed that the mixing index decreased over time owing to the segregation of large and small particles. The RNNSR demonstrated high accuracy in predicting a decrease in the mixing index associated with segregation. Next, we investigated whether the RNNSR could predict radial and axial segregation. Fig. 8a shows the simulation results of the powder mixing behavior on the cross-sections at the center of the drum from the front and side views. **Fig. 8b** and **c** show the fraction of large particles as a function of the radial and axial distances from the center of the drum, respectively. First, regarding radial segregation, as shown in Fig. 8a, large particles were observed more in the peripheral region of the powder bed, while small particles were observed more in the central core of the powder bed over time. This is a typical segregation pattern in a rotating drum [38–40]. Comparing the mixing behavior calculated by the DEM with that predicted by the RNNSR, similar segregation patterns were observed. A full movie of the segregation behavior can be found in the supplementary material. As shown in Fig. 8b, the plot near the center of the drum and drum wall showed a higher fraction of large particles owing to radial segregation. The results obtained from RNNSR were in good agreement with those obtained from the DEM. Regarding axial segregation, large

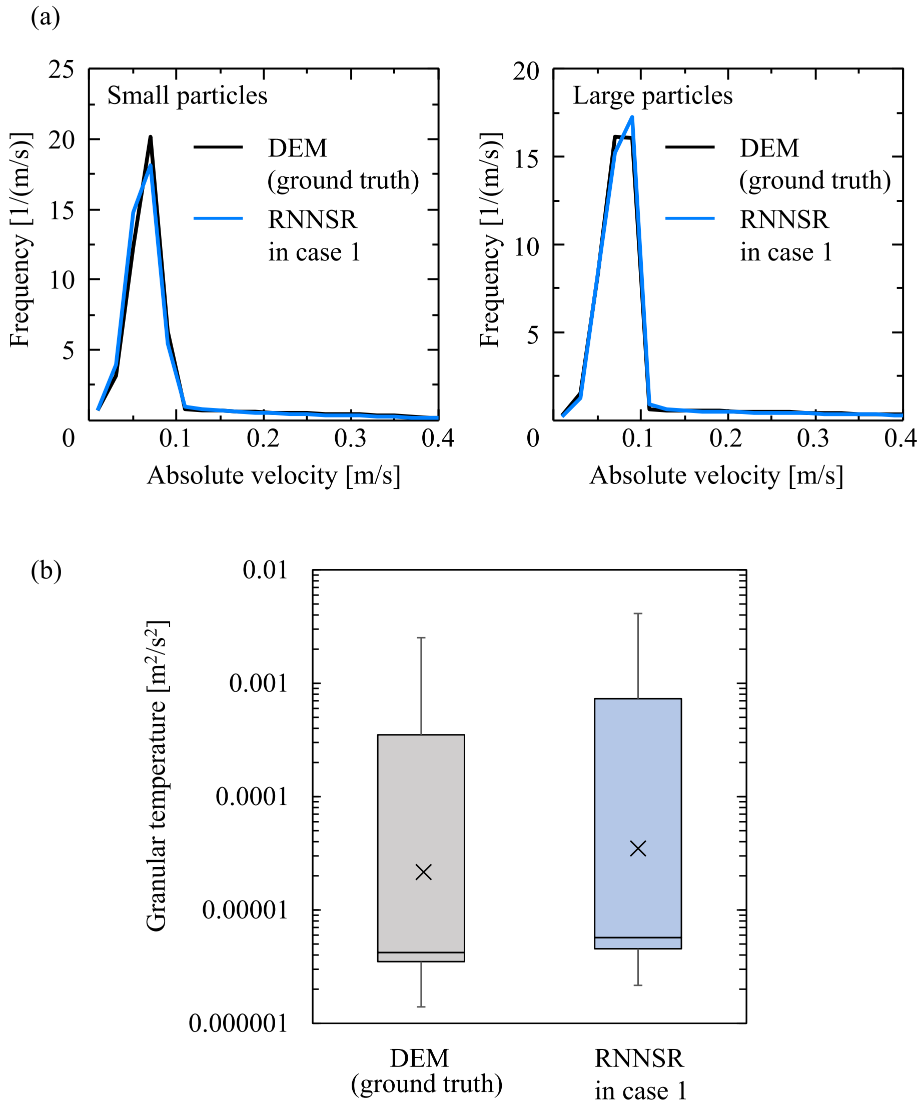


Fig. 9. (a) Distributions of large and small particle velocity calculated by the DEM and RNNSR. (b) Boxplots of the granular temperature calculated by the DEM and RNNSR. The particle velocity distribution and granular temperature were the results at 10 s. The data initiated from the RM-IC were employed in both the surrogate modeling and prediction step in the RNNSR.

particles were observed near the end walls. As shown in Fig. 8c, the fraction of large particles near the end walls increased over time. This axial segregation was accurately predicted by the RNNSR. In summary, the RNNSR in case 1 accurately predicted the segregation in terms of the mixing index and radial/axial particle fractions.

Fig. 9a shows the velocity distributions of large and small particles at 10 s obtained from the DEM and RNNSR in case 1. The results indicated that the velocity distributions obtained from the DEM and RNNSR were in good agreement for both large and small particles. This suggests that the RNNSR can accurately predict changes in the velocity distribution of large and small particles during segregation. Fig. 9b shows the calculated granular temperature, which is an indicator of the velocity fluctuation in local space [42,43], obtained from the DEM and RNNSR simulations at 10 s. In this study, the drum was divided into cubic cells, and the granular temperatures in these cells were calculated. The length of one side of the cubic cell was set to 0.04*D*, which was the same size used to calculate Lacey's mixing index. The results indicated that the granular temperatures obtained from the DEM and RNNSR were almost identical. This indicates that RNNSR can accurately predict the randomness of particle velocities. From these results, it was concluded that the RNNSR could accurately predict both the Lagrangian velocity

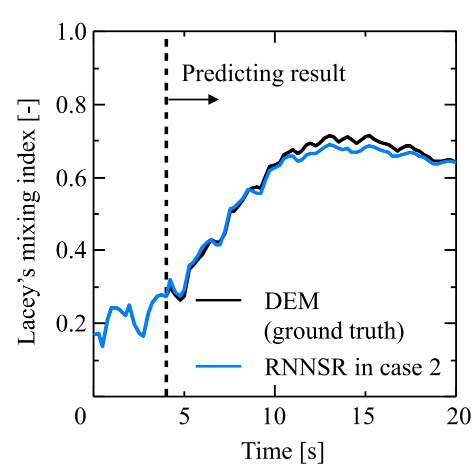


Fig. 10. Temporal changes in the degree of the particle mixing in the cases of the DEM, and RNNSR in case 2. The surrogate modeling employed the data initiated from the RM-IC, and prediction step employed the data initiated from the S-IC.

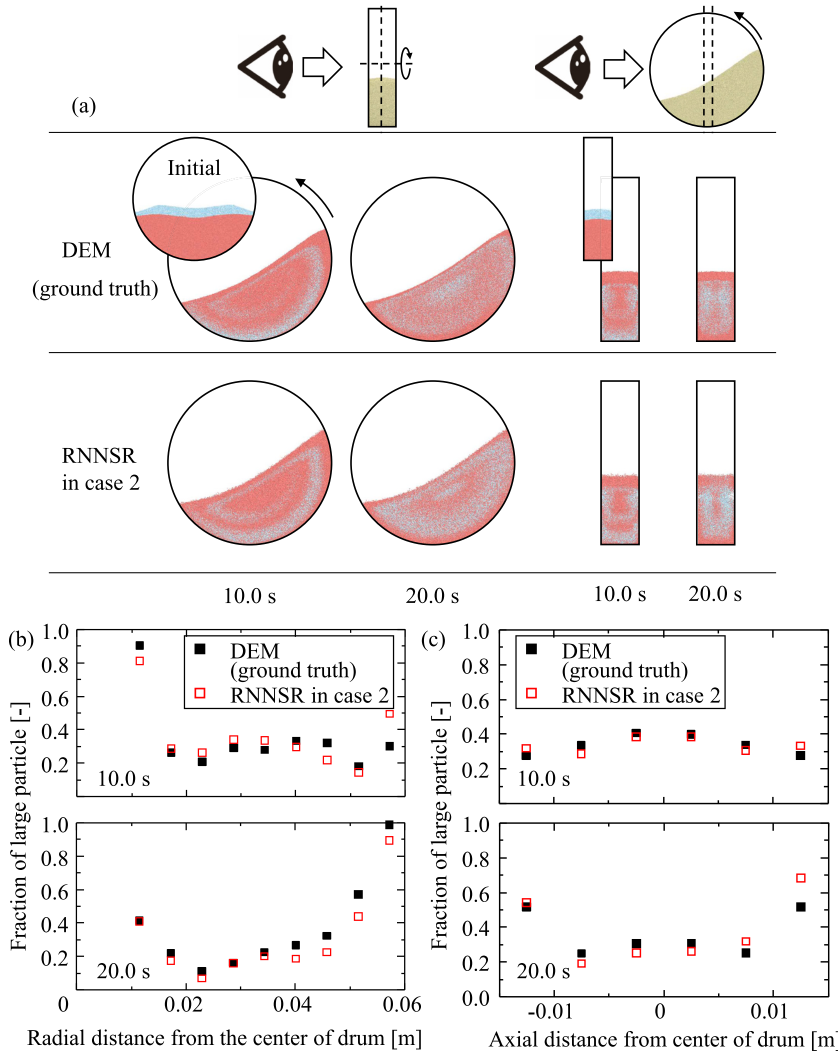


Fig. 11. (a) Powder mixing behavior on the cross-sections at the center of the drum from front and side views. (b) Fraction of large particles as a function of radial distance from the center of drum. (c) Fraction of large particles as a function of axial distance from the center of drum. The surrogate modeling employed the data initiated from the RM-IC, and prediction step employed the data initiated from the S-IC.

distribution of individual particles and the Eulerian velocity fluctuation in a local space.

4.2. Applicability of the extended-RNNSR for different initial condition

To verify the applicability of the constructed RNNSR, we predicted particle behavior under different initial conditions. Specifically, the RNNSR constructed on the data initiated from RM-IC was employed to predict the mixing and segregation behaviors initiated from S-IC (case 2 in Table 2). Fig. 10 shows the Lacey's mixing index predicted by RNNSR. The results from 4 to 20 s were predicted using the RNNSR. In the DEM, the mixing index increased until approximately 10 s and then decreased slightly. Particles initially mixed toward a homogeneous state and subsequently segregated, as observed in RM-IC. The predictions of the mixing index by RNNSR in case 2 were in good agreement with the DEM results, indicating that the RNNSR trained on data initiated from RM-IC could predict the change in the mixing index under S-IC. We then investigated mixing and segregation along the radial and axial directions. Fig. 11a shows the simulation results of the powder behaviors on the cross-sections at the center of the drum from the front and side views. First, we investigated mixing and segregation in the radial direction. The DEM simulation results (upper left of Fig. 11a) showed that the large (red) and small (cyan) particles were mixed by a typical vortex-like circulating flow in a rotating drum mixer [44,45] at 10.0 s. However, at 20.0 s, radial segregation (accumulation of large particles in the peripheral region of the powder bed) was observed. The RNNSR predicted the transient mixing behavior and radial segregation accurately. Fig. 11b shows the large particle fraction as a function of the radial distance from the center of the drum. The prediction results obtained by the RNNSR showed quantitative agreement with the DEM. Subsequently, we investigated mixing and segregation in the axial direction. The DEM simulation results (upper right of Fig. 11a) exhibited unique mixing and segregation behaviors, where large particles were observed at the center of the drum at 10.0 s, while segregation of large particles was observed in the vicinity of the end walls at 20.0 s. It should be noted that the RNNSR predicted the unique axial mixing and segregation behaviors accurately. Fig. 11c also confirms that the axial distribution of the large particle fraction was predicted by RNNSR. Consequently, it was demonstrated that the RNNSR constructed on the learning data initiated from the RM-IC was applicable to the prediction of mixing and segregation initiated from an initial particle position that differed from that of the learning data.

To understand whether the RNNSR constructed on the data initiated from the RM-IC was able to predict the mixing and segregation initiated from the S-IC, we performed surrogate modeling and prediction by exchanging the initial conditions; that is, the RNNSR constructed on the

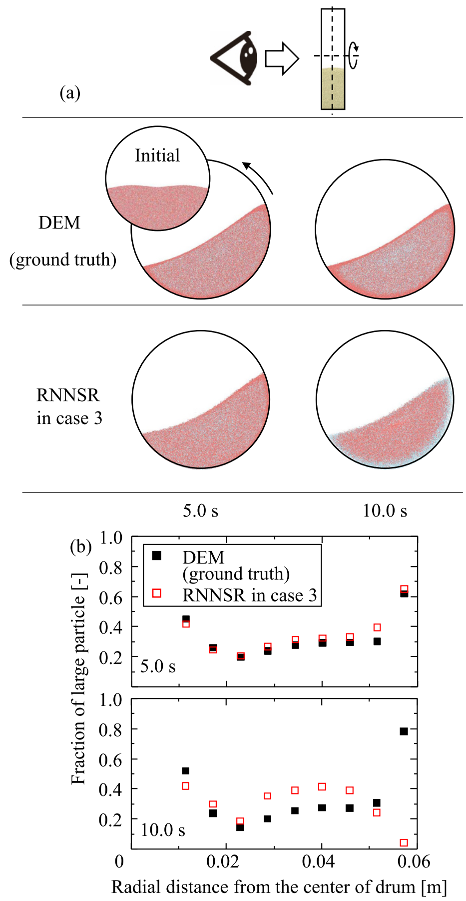


Fig. 12. (a) Powder mixing behavior on the cross-section at the center of the drum from front view. (b) Fraction of large particles as a function of radial distance from the center of drum. The surrogate modeling employed the data initiated from the S-IC, and prediction step employed the data initiated from the RM-IC.

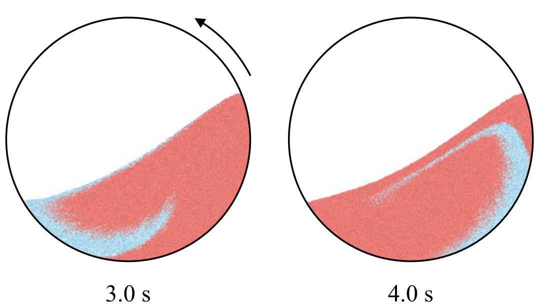


Fig. 13. The snapshots of mixing behavior in segregated initial condition at 3.0, and 4.0 s, which was used for the training data of RNNSR in case 3.

data initiated from the S-IC was employed to predict the mixing and segregation initiated from the RM-IC (case 3 in Table 2). Fig. 12 summarizes the prediction results for case 3. The mixing behavior predicted by the RNNSR exhibited the accumulation of small particles in the vicinity of the rotating cylindrical wall at 10.0 s (Fig. 12a), and this result was completely different from that in the DEM simulation result. This discrepancy is also confirmed in Fig. 12b, in which the large particle fraction predicted by RNNSR in case 3 was significantly lower at a radial distance of 0.057 m at 10.0 s. Consequently, the RNNSR constructed based on the data initiated from S-IC could not predict the mixing and segregation initiated from RM-IC. This result was attributed to the uniformity of the spatial arrangement of large and small particles in the

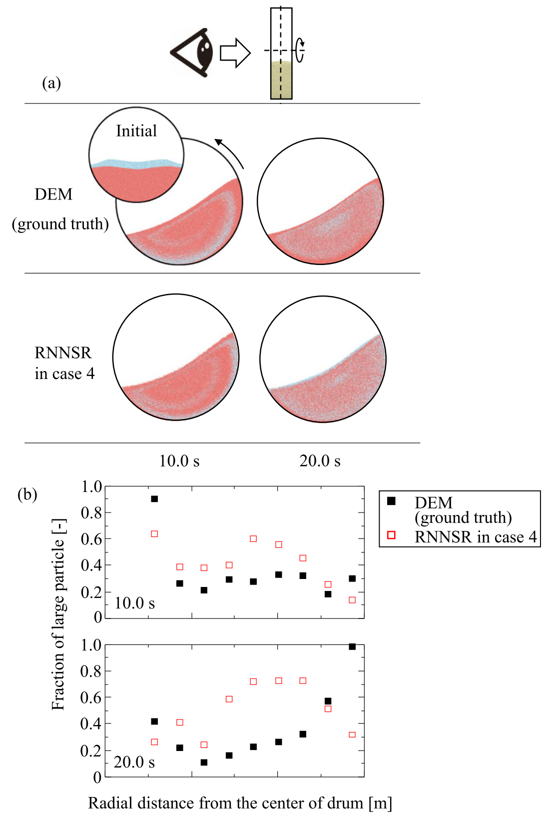


Fig. 14. (a) Powder mixing behavior on the cross-section at the center of the drum from front view. (b) Fraction of large particles as a function of radial distance from the center of drum. The data initiated from the S-IC were employed in both the surrogate modeling and prediction step in the RNNSR.

training data. The training data initiated from the S-IC consisted of the trajectories of large and small particles that were segregated (Fig. 13). These segregated data resulted in a lower prediction performance in regions where the relevant particles were absent. On the other hand, in the training data obtained from RM-IC, both large and small particles were uniformly distributed. This enabled the RNNSR to learn particle trajectories in the entire region and predict the mixing and segregation

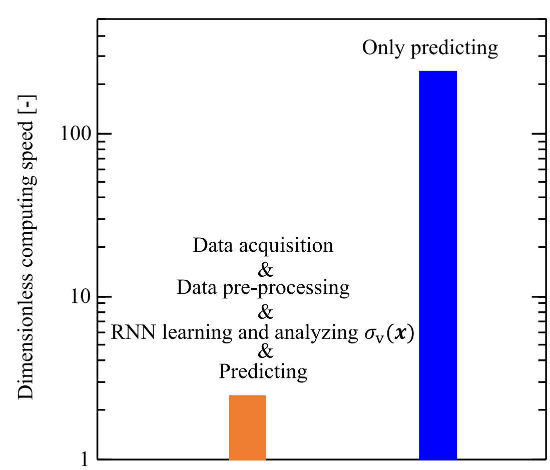


Fig. 15. Dimensionless computing speed of the extended-RNNSR in the case 1. The dimensionless computing speed was calculated for the all steps of the RNNSR (acquisition of training data, training of RNN, and prediction of particle motion) and only the prediction step, respectively. The dimensionless computing speed was defined as an effective computing speed normalized by the corresponding value in the case of the DEM.

behaviors, regardless of the spatial arrangement of large and small particles in the prediction step. For supporting this discussion, we performed the RNNSR prediction in case 4, where the data initiated from the S-IC were employed in both the surrogate modeling and prediction steps. Fig. 14 shows the prediction results for case 4. The prediction results obtained by the RNNSR in case 4 significantly deviated from those of the DEM simulation, even though the data was continuous between training and prediction. These additional results confirmed that the segregated data for the RNN training resulted in a lower prediction performance, similar to case 3. Consequently, it was found that uniformly distributed training data is necessary for accurate prediction.

Finally, we discuss the predictability characteristics acquired by the extended-RNNSR in this study. As shown in Fig. 4, in the DEM simulation results (ground truth), the locally averaged particle behavior remained unchanged even when the segregation progressed. This indicates that the physical quantities associated with the origin of segregation cannot be quantified using local averaging alone. Conversely, the extended-RNNSR effectively predicted the segregation behavior of binary-sized particles. These findings provide insights into the fact that the RNNSR did not merely acquire and predict locally averaged physical quantities but rather acquired and predicted unique particle behaviors related to the origin of segregation.

4.3. Computing speed

Fig. 15 shows the dimensionless computing speed of the extended-RNNSR in case 1 compared with that of the DEM. The dimensionless computing speed was normalized using the computing speed of the DEM. The dimensionless computing speed was calculated for all RNNSR steps (steps 1 to 4, as shown in Fig. 1) and for the prediction step (only step 4, as shown in Fig. 1). The dimensionless computing speed was evaluated by the wall clock time of a 10 s DEM and RNNSR runs under the same hardware conditions. The computing speed for the all RNNSR step was approximately 2.5 times that of the DEM, whereas it was 240 times faster for the prediction step. Most of the computing time was spent on data acquisition, resulting in a much lower computing speed for all RNNSR steps than that in prediction only. This implies that the computing speed for all the RNNSR steps can be improved if a longer target simulation time is set. In summary, it was demonstrated that RNNSR enables much faster computation than DEM, even in a binary-sized particle system.

5. Conclusion

In this study, an extended-RNNSR model was developed to simulate the mixing and segregation of powders with inhomogeneous properties. To gain insight into extending the RNNSR to binary-sized particle systems, we first conducted a DEM analysis of the Lagrangian and Eulerian behaviors of binary-sized particles during segregation in a rotating drum mixer. DEM analyses of the Lagrangian and Eulerian behaviors revealed that the Lagrangian behavior was dependent on the particle size, whereas the Eulerian behavior was independent of the particle size. Based on these insights, an extended-RNNSR for binary-sized particle systems was developed by adding particle size data to the RNN input data. The prediction results of the RNNSR showed good agreement with those of the DEM in terms of the mixing degree, particle velocity, and granular temperature. Moreover, it was demonstrated that the RNNSR constructed on the learning data initiated from the RM-IC was applicable to the prediction of mixing and segregation initiated from the S-IC, which was different from that of the learning data. The RNNSR also demonstrated a much faster computing speed than the DEM, even in binary-sized particle systems.

The findings of this study indicate that learning the particle trajectories in the entire region is very important for predicting the mixing and segregation behaviors of large and small particles. It was also found that the RNNSR not only acquired and predicted locally averaged physical quantities but also acquired and predicted unique particle behaviors related to the origin of segregation. We believe that the RNNSR can significantly contribute to the understanding of the segregation phenomenon, industrial applications of powder simulations, and digital transformation in powder technology.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.powtec.2025.120811.

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CRediT authorship contribution statement

Naoki Kishida: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hideya Nakamura:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuji Ohsaki:** Validation, Resources. **Satoru Watano:** Validation, Resources.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that might have influenced the work reported in this study.

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Appendix A. Governing equations of DEM

The individual particle motion was calculated based on Newton’s equation of motion. The equations of motion for translation and rotation are as follows:

$$m\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t_{\mathrm{DEM}}} = \boldsymbol{F}_{\mathrm{cn}} + \boldsymbol{F}_{\mathrm{ct}} + m\boldsymbol{g} \tag{A-1}$$

$$I\frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t_{\mathrm{DEM}}} = \boldsymbol{M} + \boldsymbol{M}_{\mathrm{rf}} \tag{A-2}$$

where m, $\boldsymbol{v}$, t_{DEM}, $\boldsymbol{F}_{\mathrm{cn}}$, $\boldsymbol{F}_{\mathrm{ct}}$, and $\boldsymbol{g}$ are the particle mass, particle velocity, time step in the DEM, normal contact force, tangential contact force, and gravity, respectively. I, ω, $\boldsymbol{M}$, and $\boldsymbol{M}_{\mathrm{rf}}$ are the moment of inertia, the angular velocity of the particle, torque due to the tangential contact

force, and torque due to the rolling resistance, respectively. The contact forces were calculated using the Hertz-Mindlin contact force model [46–48]:

$$\boldsymbol{F}_{\mathrm{cn}} = -k_{\mathrm{n}}\delta_{\mathrm{n}}\widehat{\boldsymbol{n}} - \eta_{\mathrm{n}}\boldsymbol{v}_{\mathrm{r,n}} \tag{A-3}$$

$$\boldsymbol{F}_{\mathrm{ct}} = \begin{cases} -k_{\mathrm{t}}\delta_{\mathrm{t}}\widehat{\boldsymbol{t}} - \eta_{\mathrm{t}}\boldsymbol{v}_{\mathrm{r,t}}\left(\text{if } \left| -k_{\mathrm{t}}\delta_{\mathrm{t}}\widehat{\boldsymbol{n}} - \eta_{\mathrm{t}}\boldsymbol{v}_{\mathrm{r,t}}\right| < \mu|\boldsymbol{F}_{\mathrm{n}}|\right) \\ -\mu|\boldsymbol{F}_{\mathrm{n}}|\widehat{\boldsymbol{t}}\left(\text{if } \left| -k_{\mathrm{t}}\delta_{\mathrm{t}}\widehat{\boldsymbol{n}} - \eta_{\mathrm{t}}\boldsymbol{v}_{\mathrm{r,t}}\right| \geq \mu|\boldsymbol{F}_{\mathrm{n}}|\right) \end{cases} \tag{A-4}$$

where k, δ, η, $\boldsymbol{v}_{\mathrm{r}}$, and μ represent the stiffness, overlap, damping stiffness, relative velocity, and friction coefficient, respectively. Subscripts n and t denote the normal and tangential directions, respectively. $\widehat{\boldsymbol{n}}$ and $\widehat{\boldsymbol{t}}$ denote the normal and tangential unit vectors at the contact point, respectively. The stiffness and damping coefficients are expressed by the equations listed in Table A-1.

Table A-1
Stiffness and damping coefficients of the Hertz–Mindlin contact model.

	Normal direction			Tangential direction
Stiffness	$k_{\mathrm{n}} = \frac{4}{3}Y^* \sqrt{r^*\delta_{\mathrm{n}}}$			$k_{\mathrm{t}} = 8G^* \sqrt{r^*\delta_{\mathrm{n}}}$
Damping coefficient	$\eta_{\mathrm{n}} = 2\sqrt{\frac{5}{6}}B\sqrt{c_{\mathrm{n}}m^*}$			$\eta_{\mathrm{t}} = 2\sqrt{\frac{5}{6}}B\sqrt{k_{\mathrm{t}}m^*}$
$c_{\mathrm{n}} = 2Y^*\sqrt{r^*\delta_{\mathrm{n}}}$	$r^* = \frac{r_ir_j}{r_i + r_j}$			$m^* = \frac{m_im_j}{m_i + m_j}$
$B = \left\{1 + \left(\frac{\pi}{\ln e}\right)^2\right\}^{-1/2}$	$\frac{1}{Y^*} = \frac{1-\nu_i^2}{Y_i} + \frac{1-\nu_j^2}{Y_j}$			$\frac{1}{G^*} = \frac{2-\nu_i}{G_i} + \frac{2-\nu_j}{G_j}$
Y^*	Reduced Young's modulus [Pa]	G	Shear modulus [Pa]	
Y	Young's modulus [Pa]	m^*	Reduced particle mass [kg]	
r^*	Reduced particle radius [m]	m_i, m_j	Particle mass [kg]	
r_i, r_j	Particle radius [m]	e	Restitution coefficient [−]	
G^*	Reduced shear modulus [Pa]	ν_i, ν_j	Poisson's ratio [−]	

The torques due to the tangential contact force and rolling friction are described as follows:

$$\boldsymbol{M} = r\widehat{\boldsymbol{n}} \times \boldsymbol{F}_{\mathrm{ct}} \tag{A-5}$$

$$\boldsymbol{M}_{\mathrm{rf}} = -\mu_{\mathrm{rf}}r|\boldsymbol{F}_{\mathrm{cn}}|\frac{\boldsymbol{\omega}}{|\boldsymbol{\omega}|} \tag{A-6}$$

where r and μ_{rf} are the particle radius and rolling friction coefficient, respectively.

Data availability

Data will be made available on request.

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